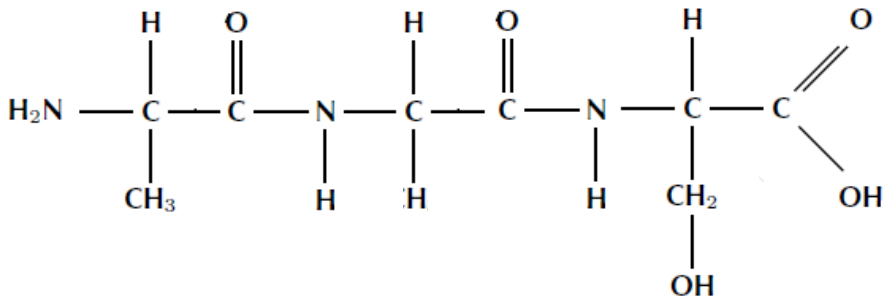
Answers to Section Multiple Choice Questions:
Answers to Proteins, Fats and Carbohydrate Past SAC (Chapters 15 and 16) Questions

Question 1

The structure of a molecule is represented below:



Which of the amino acids listed would not be produced when this molecule undergoes hydrolysis?

A. Alanine
B. Glycine
C. Threonine
D. Serine

Answer is C: This is a tripeptide made up of 3 amino acids: alanine, glycine and serine.

Question 2

Which of the following fatty acids is the most saturated?

A. Linoleic 2 x C=C
B. Palmitoleic 1 x C=C
C. Oleic 1 x C=C
D. Myristic 0 x C=C

Answer is D: Most saturated = least C=C double bonds.

Question 3

Amino acids are classified as essential or non-essential. Which of the following statements is **true**?

A. The Z-groups on essential amino acids are always polar.
B. Essential amino acids cannot be manufactured in the body and must be included in the d
C. Non-essential amino acids do not form bonds within the tertiary structure of a protein.
D. Non-essential amino acids are obtained through the foods we eat.

Answer is B: All other answers are false.

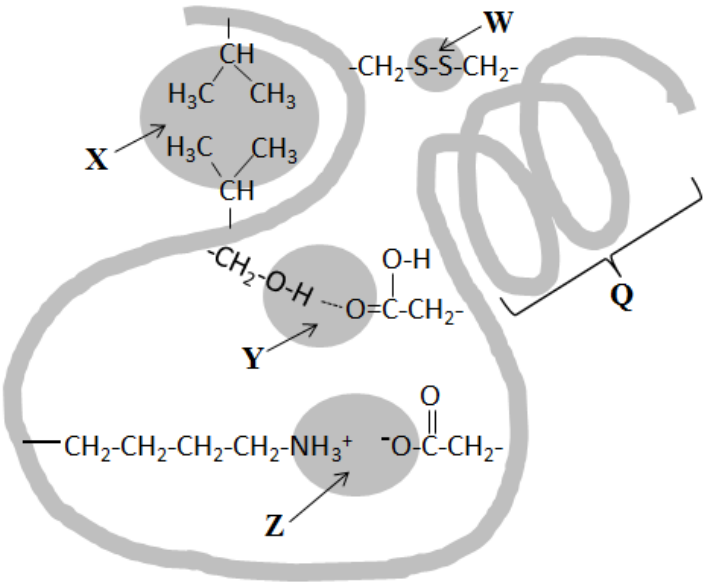
Question 4

Select the best definition for an enzyme.

A. An enzyme is an amino acid that speeds up chemical reactions.
B. An enzyme is a non-specific organic catalyst that speeds up chemical reactions.
C. Enzymes are selective proteins that speed up chemical reactions.
D. Enzymes are inorganic and speed up chemical reactions using the lock and key m

Answer is C: All other answers are false.

Use the following information to answer Questions 5 - 7
The following diagram is a simplified representation of a section of a polymer.



Question 5
The types of bonding represented at X , Y and Z are:

	X	Y	Z
A.	dipole-dipole force	ionic bond	peptide link
B.	peptide link	dipole-dipole force	hydrogen bond
C.	dispersion forces	hydrogen bond	ionic bond
D.	dispersion forces	dispersion forcess	ionic bond

Answer is C

Question 6
The type of bonding represented at W, and the amino acids that were involved in the bonding are:

	bonding	amino acid 1	amino acid 2
A.	disulphide link	cysteine	cysteine
B.	glycosidic link	cysteine	cysteine
C.	ether link	cysteine	methionine
D.	peptide link	cysteine	methionine

Answer is A

Question 7

The section of polymer represented by Q forms in which structure, and is held together by which type of bonding?

	structure	type of bonding
A.	secondary	covalent bond
B.	secondary	hydrogen bond
C.	tertiary	covalent bond
D.	tertiary	hydrogen bond

Answer is B: Hydrogen bonds form between the C=O group of one peptide link and the N-H of another peptide link.

Question 8

A fatty acid molecule contains 18 carbon atoms and 2 C=C double bonds. The fatty acid could be

A. $C_{18}H_{32}O_2$
B. Oleic acid
C. $C_{18}H_{36}O_2$
D. $C_{18}H_{32}COOH$

Answer is A

Question 9

A single strand of DNA contains 10 nucleotides. There are 24 hydrogen bonds between the strand and its complementary strand. The number of adenine **and** thymine bases in the strand must be.

A. 4 = 8 x H-bonds (would require 5.3 G&C)
B. 5 = 10 x H-bonds (would require 4.6 G&C)
C. 6 = 12 x H-bonds (would require 4 G&C)
D. 7 = 14 x H-bonds (would require 3.3 G&C)

Answer is C: 2 x H-bonds between A&T, and 3 x H-bonds between C&G.

Question 10

If an enzyme is produced from 1956 amino acid monomers, how many water molecules would be released?

A. 1956
B. 1014
C. 1955
D. 3912

Answer is C: (n – 1) One less H_2O molecule than monomers.

Question 11

A nucleotide consists of

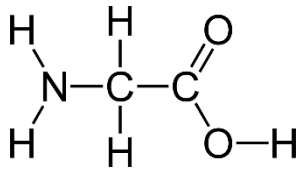
A. adenine, guanine, thymine, and cytosine
B. a purine base, a phosphate group, adenine and guanine
C. a deoxyribose sugar, a base, and a nitrate group
D. a phosphate group, a deoxyribose sugar group, and a purine base

Answer is D

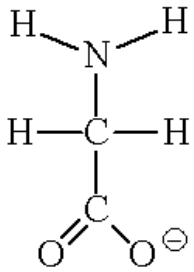
Question 12

At pH 6, the amino acid glycine exists in its **zwitterion form**. Which of the following structures would represent glycine as it exists at pH 3?

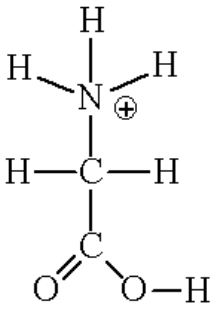
A.

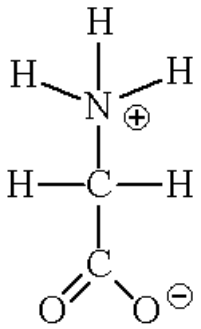


C.



B.





Answer is B:
pH3 (more acidic than zwitterion pH), so the molecule would accept H^+

Question 13

The molecular formula of alanine is $C_3H_7O_2N$. If three alanine molecules react together to form a tripeptide, the relative molecular mass of this tripeptide would be:

A. 89
B. 231
C. 267
D. 283

Answer is B:

$3 \times M(C_3H_7O_2N) - 2 \times M(H_2O)$

$= (3 \times 89) - (2 \times 18)$

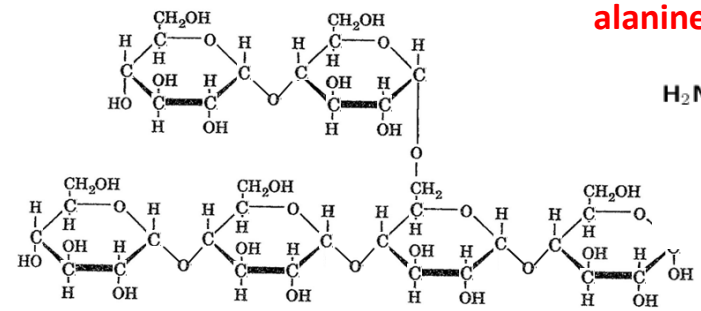
$= 231$

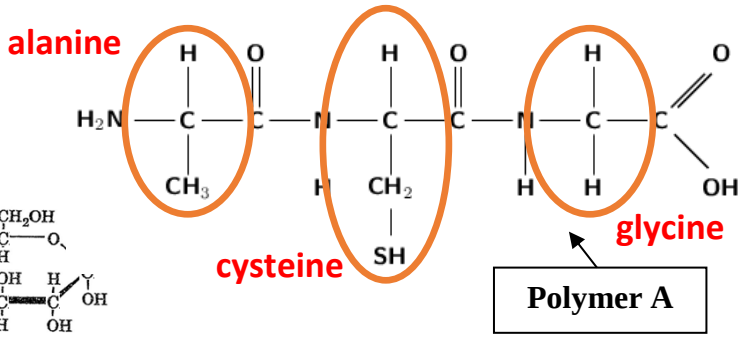
Section B Short Answer Questions:

Proteins, Fats and Carbohydrate Past SAC (Chapters 15 and 16) Questions

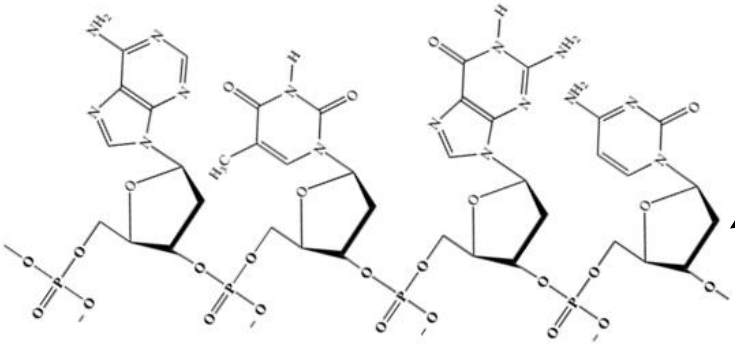
Question 1

Many of the biomolecules found in our body are polymers. The structures of three such polymers are shown below:



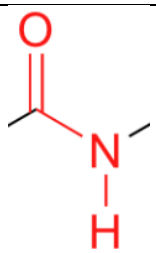
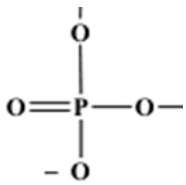


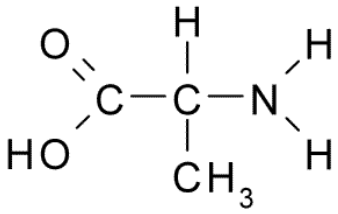
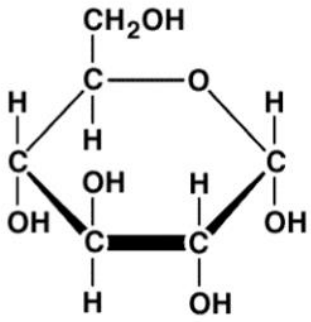
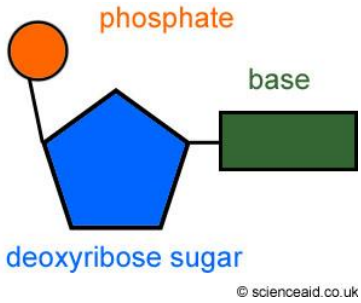
Polymer B



Polymer C

a. Use polymers A, B, and C above to complete the following table:

	Polymer A	**Polymer B**	**Polymer C**
Type of biomolecule	Protein	Carbohydrate (Starch) Or polysaccharide	DNA Or polynucleotide
Name of link	Peptide link Or amide link	Ether link Or Glycosidic link	Phosphodiester bond
Structure of link		-O-	

Number of monomers	**3**	6	**4**
Structure and name of **one** monomer.	**Full** structure:  Name: (Must match structure) **Alanine** (shown), **cysteine, or glycine**	**Full** structure:  Name: **glucose**	Use **symbols/words** to represent structure:  Name: **nucleotide**

11 marks (1 mark per box)

b. What type of reaction occurs between the monomers in order to produce polymer C?

Condensation reaction (1 mark)

1 mark

c. Proteins and DNA both have secondary structures maintained by hydrogen bonding. The strength of bonding that results from the hydrogen bonds throughout the secondary structure of a protein is fairly consistent. This is not the case for DNA. Explain.

In proteins one H-bond exists between the C=O group on one peptide link and the

N-H group in another peptide link between amino acid monomers (1 mark)

In DNA there are 3 H-bonds between G&C and 2 H-bonds between A&T (1 mark)

The protein H-bonds form a consistent pattern while the DNA strand sequence

will vary (1 mark)

3 marks

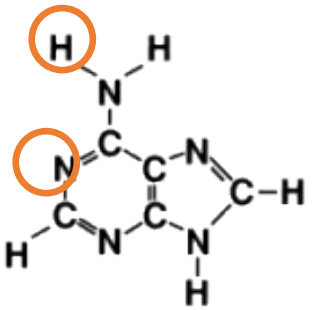
Total 15 marks

Question 2

Molecule X forms a component of DNA. It forms specific bonds with another nitrogen containing compound on a complimentary strand of DNA.

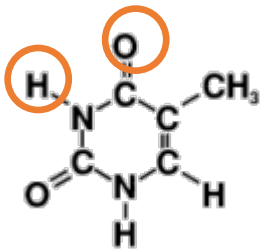
a. **Circle** the components on Molecule X that are directly involved in maintaining the **secondary structure** of DNA.

Molecule X



1 mark

b. Draw the structure of the base which is complimentary to Molecule X and circle the components of this molecule that are directly involved in maintaining the secondary structure of DNA with Molecule X.

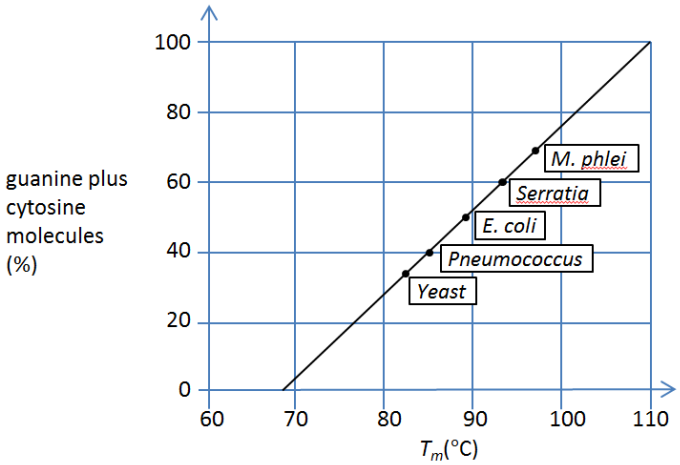


(Thymine – semistructural abbreviations ok)

If students circle both groups instead of individual atoms only take 1 mark off

2 marks

c. When double-stranded DNA is heated slowly, the double helix unravels to a random coil configuration over a narrow temperature range. The midpoint temperature of this process is known as the melting temperature (*Tm*). The graph below shows the *Tm* of various samples of DNA (taken from different living things) as a function of the base composition.



i. With reference to the structure of DNA, explain the trend depicted in the graph.

3 H-bonds between each guanine and cytosine pair and 2 H-bonds between each thymine and adenine pair in DNA (1 mark)

T_m increases for each strand with a higher proportion of G-C pairs as more energy is required to break the increased number of H-bonds (1 mark)

ii. In a sample of double stranded DNA from *E. coli* which contains 3000 base pairs, how many adenine and thymine molecules in total wold be present in a single strand of its DNA?

***E.Coli* has 50% C-G pairs and therefore 50% of A-T pairs** (1 mark)

Therefore 1500 A-T base pairs in a double strand and 1500 A+T molecules in a single strand of DNA (1 mark)

2 + 2 = 4 marks

d. In a cell, DNA strands are bonded tightly to proteins known as histones. These proteins contain a high proportion of the amino acids lysine and arginine. In the acidic environment of the DNA molecule, the side chain groups of these amino acids are charged.
Name the type of bonding that exists between histones and the DNA molecule, and explain how this bonding occurs between them.

Ionic bonding (1 mark)

DNA backbone is negatively charged due to phosphate groups and side chain groups of histone amino acids are positive allowing for ionic bonding to occur

(1 mark)

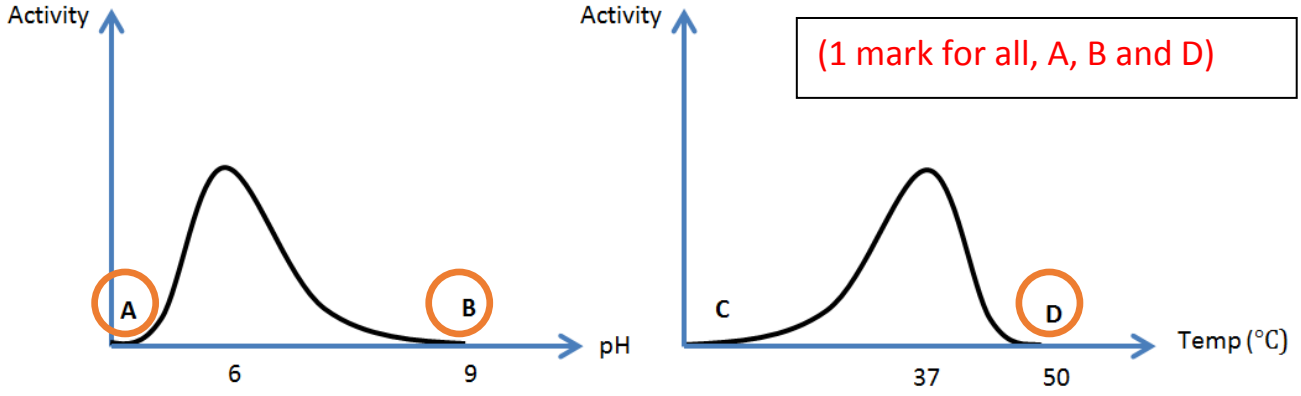
2 marks

Total 9 marks

Question 3

a. The graphs below illustrate the activity of an enzyme as a function of pH and temperature.

i. In which region of **each** of the graphs has protein denaturation occurred? Circle the appropriate answer(s) A, B, C, or D below.



(1 mark for all, A, B and D)

ii. Explain why the catalytic function of an enzyme molecule is dependent on the way the molecule is folded.

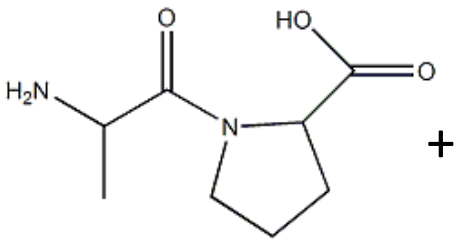
The specific folding allows for interactions to occur between the enzyme and the substrate which result in the breaking of bonds in the substrate. (1 mark)

If the folding is in any way distorted, the interactions will no longer align and the enzyme can no longer catalyse the reaction. (1 mark)

1 + 2 = 3 marks

b. Using full structural formulae, showing all bonds, write an equation to represent the reaction between alanine and proline to form the dipeptide ala-pro.

Ala (structure) + pro (structure) →



+ H_2O

(1 mark for equation, 1 mark for all bonds shown (for ala, pro and dipeptide, do not penalise if H_2O is not shown as a full structure)

2 marks

Total 5 marks

Calorimetry Past SAC Questions

Question 4

The electrical calibration data for a solution calorimeter is shown below.

Volume of solution	**100.0 mL**
Initial temperature	**18.27°C**
Voltage	**6.10 V**
Current	**2.13 A**
Time	**10.0 minutes**
Final Temperature	**32.93°C**

a) Determine the calibration factor for this calorimeter

E = V x I x t = 6.10 x 2.13 x (10 x 60) = 7796 J [1 mark]

ΔT = 32.93 – 18.27 = 14.66°C

Calibration factor = $\frac{7796}{14.66}$ = 532 J°C^{-1} [1 mark]

b) When 0.370 g of zinc powder was added to 100.0 mL of 1.0 M aqueous copper (II) sulfate solution in the calorimeter, a 2.32°C temperature increase occurred.

(i) Write an ionic equation for the reaction between the zinc and the copper (II) solution.

$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ **[1 mark]**

(ii) Calculate the change in enthalpy, ΔH, for this reaction.

n(Zn) = m/M = 0.370 g/65.4 = 5.66 x 10^{-3} mol

n(Cu^{2+}) = c x V = 1.0 x 0.1 = 0.10 mol

Zn is the limiting reactant [1 mark]

E = CF x ΔT = 532 x 2.32 = 1.23 x 10^{3} J = 1.23 kJ [1 mark]

5.66 x 10^{-3} mol of Zn releases 1.23 kJ

Hence 1.0 mol of Zn releases $\frac{1.23}{5.66 \times 10^{-3}}$ = 217 kJ

ΔH = -217 kJ mol^{-1} (since the reaction is exothermic) [1 mark] and [1 mark for correct number of significant figures]

4 marks
Total 7 marks